INCIDENT REPORT

Enterprise SOC Detection Lab — Ransomware Simulation & Detection Engineering

Reference: IR-2026-001 | **Classification:** LAB / SIMULATED | **Severity:** CRITICAL | **Date:** July 26, 2026 | **Analyst:** Idris Lawan Mustapha

1. Executive Summary

This report documents a simulated ransomware attack executed against an isolated Windows 10 endpoint in a controlled VMware lab environment. The objective was twofold: to replicate the tactics, techniques, and procedures (TTPs) used by modern ransomware operators, and to evaluate the detection capability of the Wazuh SIEM platform against these techniques.

The attack chain spanned seven distinct MITRE ATT&CK techniques — from initial reconnaissance through ransomware recovery inhibition. Wazuh's default detection rules successfully logged process creation events throughout the attack, but critically failed to classify the most dangerous ransomware-staging techniques at appropriate severity levels. The `vssadmin` shadow copy deletion command — a technique used universally by ransomware strains including LockBit, BlackCat, and WannaCry — was classified as a low-level generic event (Level 3), meaning it would be invisible to a standard SOC analyst monitoring a live dashboard.

Following this gap analysis, three custom XML detection rules were engineered and deployed to the Wazuh Manager. Upon re-execution of the attack chain, all three rules fired immediately — escalating the critical ransomware-staging techniques to **Level 12 Critical** alerts with MITRE ATT&CK T1490 mapping. This demonstrates the core value of Detection Engineering as a discipline distinct from, and complementary to, standard SOC analysis.

2. Lab Architecture & Environment Blueprint

The lab consists of three virtual machines operating within an isolated VMware host-only network segment. No external internet traffic is permitted from the victim or attacker nodes during simulation.

Role	OS / Version	IP Address	Key Tools / Telemetry	Purpose
Attacker Node	Kali Linux 2026.1	`192.168.184.129`	Nmap 7.99, Metasploit v6.4	Simulate threat actor execution
Victim Endpoint	Windows 10 21H1 (Build 19043)	`192.168.184.166`	Sysmon v15.21, Wazuh Agent v4.14.5	Target endpoint under attack
SIEM Manager	Wazuh Server (Ubuntu OVA)	`192.168.184.130`	Wazuh Manager & Dashboard v4.14.5	Log collection, detection & alerting

3. Attack Timeline (Verified Evidence Timestamps)

Time (EDT/UTC)	Event Description	Executed Command	MITRE ATT&CK
03:54:00 EDT	Nmap reconnaissance scan launched from Kali against victim endpoint.	`nmap -sV -O -p- 192.168.184.166`	T1595
04:07:00 EDT	Msfvenom payload generated on Kali; hosted via Python HTTP server on port 8080.	`msfvenom -p windows/x64/meterpreter/reverse_tcp`	T1204.002
04:37:00 EDT	Multi/handler listener started on Kali (port 4444).	`msfconsole` (multi/handler)	T1571
04:47:59 EDT	Payload executed on victim endpoint. Reverse TCP connection established.	`payload.exe`	T1571
04:50:00 EDT	Privilege escalation executed via `getsystem` (Named Pipe Impersonation).	`getsystem`	T1134
04:50:30 EDT	Credential dumping via `hashdump` extracting local SAM database hashes.	`hashdump`	T1003
05:00:00 EDT	Recovery inhibition executed: Volume Shadow Copies deleted.	`vssadmin delete shadows /all /quiet`	T1490
05:00:15 EDT	Windows Backup catalog deleted.	`wbadmin delete catalog -quiet`	T1490
09:04:00 EDT	Defender Real-Time Protection disabled programmatically.	`Set-MpPreference -DisableRealtimeMonitoring $true`	T1562.001
09:10:00 EDT**	Payload binary downloaded to endpoint disk.	`Invoke-WebRequest -Uri http://.../payload.exe`	T1105
10:14:00 EDT	SIEM Gap Analysis conducted — query for `vssadmin` yielded No Results in default alerts.	Wazuh Discover Search	Gap Analysis
13:05:59 UTC	Custom Rule 100001 fired at Level 12 Critical upon attack re-execution.	Wazuh Alert Engine	T1490

4. MITRE ATT&CK Technique Mapping

ID	Tactic	Technique	Tool Used
T1595	Reconnaissance	Active Scanning	Nmap 7.99
T1204.002	Execution	User Execution: Malicious File	Msfvenom Payload
T1571	Command & Control	Non-Standard Port	Metasploit Reverse TCP (4444)

ID	Tactic	Technique	Tool Used
T1134	Privilege Escalation	Access Token Manipulation	getsystem (Named Pipe)
T1003	Credential Access	OS Credential Dumping	hashdump (SAM Extraction)
T1562.001	Defense Evasion	Impair Defenses: Disable Tools	PowerShell Set-MpPreference
T1490	Impact	Inhibit System Recovery	vssadmin, wbadmin, bcdedit
T1486	Impact	Data Encrypted for Impact	Simulated Payload Drop

5. Indicators of Compromise (IOCs)

- **Attacker IP:** `192.168.184.129` (Kali Linux)
- **Victim IP & MAC:** `192.168.184.166` | MAC: `00:0C:29:66:5D:2A`
- **C2 Channel:** Reverse TCP Socket on Port `4444`
- **Malicious Binary:** `C:\Users\Public\payload.exe` (Size: 7,680 bytes)
- **Artifact Files:** `C:\Users\Public\Desktop\README_DECRYPT.txt`, `C:\Users\Public\Desktop\FINANCIAL_2026.xlsx.enc`
- **Monitored Command Triggers:**
  - `vssadmin.exe delete shadows /all /quiet`
  - `wbadmin.exe delete catalog -quiet`
  - `bcdedit.exe /set {default} recoveryenabled No`
  - `powershell.exe Set-MpPreference -DisableRealtimeMonitoring $true`

6. Detection Gap Analysis & Custom Rule Engineering

Default Wazuh rules failed to raise alerts above Level 3 for `vssadmin delete shadows`. Three custom detection rules were engineered targeting Sysmon Event ID 1 (`61603`) process creation logs in `/var/ossec/etc/rules/local_rules.xml` :

Rule ID	Severity Level	Trigger Command	Description
100001	**Level 12 CRITICAL**	`vssadmin delete shadows`	Ransomware Staging: Volume Shadow Copy Deletion (T1490)
100002	**Level 11 HIGH**	`bcdedit recoveryenabled No`	Ransomware Staging: Windows Boot Recovery Disabled (T1490)
100003	**Level 11 HIGH**	`wbadmin delete catalog`	Ransomware Staging: Backup Catalog Deletion (T1490)

7. Recommendations & Remediation

- **Immediate Action:** Deploy custom T1490 rules (100001–100003) across all production endpoints.

- **Access Restriction:** Restrict `vssadmin.exe` and `wbadmin.exe` execution via AppLocker / WDAC policies.
- **Automated Response:** Configure Wazuh Active Response to automatically isolate endpoints triggering Level 12 alerts.

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